

R Sarang

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Education

Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Kancheepuram *May 2028*

Dual Degree: B.Tech in Electronics & Communication and M.Tech (Honors) in **VLSI Design** *CGPA: 9.01*

Relevant Coursework: **Digital IC Design, CMOS VLSI System Design, VLSI Testing & Testable Design (DFT), Microprocessors**

Indian Institute of Technology (IIT), Madras (Online) *May 2028*

BS in **Data Science and Applications** *CGPA: 8.48*

Relevant Coursework: Linear Algebra, Optimization, Probabilistic Models, Machine Learning

Technical Skills

Hardware & VLSI: Verilog, SystemVerilog, RTL Design, Digital ASIC Design, Design for Test (DFT), Fault Detection, CPU Design, RISC-V Architecture, CMOS Logic.

EDA Tools & Simulation: QuestaSim, Xilinx Vivado, Intel Quartus Prime, Cadence (Genus, Conformal, Innovus, Virtuoso).

Software & Scripting: C, C++, Python, Tcl, Bash/Shell Scripting, Linux (Ubuntu), Git, **Object-Oriented Programming**.

Embedded Systems: STM32, ESP32, Raspberry Pi, NVIDIA Jetson, Sensor Integration (ADC/DAC, I2C, SPI).

Certifications & Awards

- ISWDP Certification** – Advanced training in semiconductor process technology, device design, and microelectronics R&D tools *2026*
- Highest CGPA Award (Ranked 1st in VLSI)** – IIITDM Kancheepuram *2023-24, 2024-25*
- Hardware & Autonomy Awards:** 1st Place at AtomQuest (2025) | Judges Innovation Award, Caterpillar Autonomy Challenge, IITM (2026)

Core Hardware & VLSI Projects

1 TOPS (1 Tape Out Per Student) – VLSI Society of India (VSI) & C-DAC *Nov 2025 – Present*

- Selected for a highly competitive national cohort to design and tape-out a **22nm TSMC RISC-V System-on-Chip (SoC)** targeted for assistive audio signal processing, integrating a **RISC-V (Ibex) CPU core** over a custom **AHB interconnect**.
- Digital ASIC / RTL design** of a dedicated **256-point FFT/IFFT hardware accelerator** in Verilog/SystemVerilog for real-time frequency-domain noise suppression, using a resource-shared (folded) butterfly datapath over 16-bit fixed-point complex arithmetic to minimize silicon area.
- Architected a **zero-copy, ping-pong RAM datapath** enabling the FFT, IFFT, and CPU filtration stages to operate in-place on shared SRAM, cutting originally planned DMA transfers by **~80%**, backed by a round-robin **AHB arbiter/decoder** with sticky-grant arbitration across CPU and DMA masters.
- Verified full-chip functionality via an **800M-cycle (100 MHz) SystemVerilog testbench** exercising four consecutive real-time audio frames end-to-end, with in-sim cycle-count instrumentation to benchmark CPU vs. DMA vs. accelerator performance.

Custom RISC-V CPU on FPGA – eYRC 2025 (MazeSolver) *Sep 2025 – Mar 2026*

- Designed and verified a custom **32-bit RISC-V CPU core** from scratch using **Verilog (CPU Design, RTL Design)**.
- Synthesized and implemented the RTL on an **Intel Quartus FPGA**, achieving a maximum clock frequency of **150 MHz**.
- Engineered hardware-level control units to handle real-time I/O sensor processing, validating closed-loop PID motor actuation on an isolated single-motor testbench with zero OS overhead.

Experience

Controls Engineering Intern – Kineshia Robotics *May 2026 – Aug 2026*

- Engineered prehensile walking controllers and complex null-space projection algorithms for the Graspman open-chain robotic manipulator using **C++/Python (Object-Oriented Programming)**, increasing operational workspace and grasping success rates by 20%.
- Executed system-level hardware planning, developing 50+ comprehensive CAD models and finalizing the Bill of Materials (BoM), accelerating physical prototyping cycles by 3 weeks.
- Optimized Prioritized Inverse Kinematics (IK) control loops, reducing computational latency by 40% (to under 1ms) to ensure real-time stability and high-frequency (1 kHz) trajectory execution.

Leadership & Collaboration

Student Technical Manager & Co-Lead – MaRS (Mars Rover Students) Club *2025-2026*

- Led technical hardware integration and mentored a multidisciplinary engineering team of **40** students.
- Presented system architecture and embedded design workflows to **NASA JPL scientists** at IEEE WAMS 2025.